

Insider Threat Scenarios

Based on CERT Dataset Features

Section 1 — Logon Scenarios

#	Scenario Name	Description	Key Features	Label	Severity
1	After-Hours Midnight Access	Employee logs in at 01:30 AM, accesses internal systems for 2+ hours	after_hours_logon=1, logon_count HIGH, session_duration_avg HIGH	SUSPICIOUS	HIGH
2	Multi-PC Lateral Movement	Single user authenticates on 6 different machines within 30 minutes — physically impossible	unique_pc_count=6, logon_count=6, remote_logon_count HIGH	INSIDER	CRITICAL
3	Normal Daily Login	Employee logs in at 9:00 AM, uses assigned laptop all day, logs off at 6:00 PM	logon_count=1, after_hours_logon=0, weekend_logon=0, session_duration_avg=9	LEGITIMATE	LOW
4	Terminated Employee Active Session	User account still active 3 days after HR submitted termination; logged in from remote IP	after_hours_logon HIGH, remote_logon_count HIGH, logon_count HIGH	INSIDER	CRITICAL
5	Short Repeated Sessions	User logs in/off 15 times in one day, each session only 3-5 minutes — data extraction loops	logon_count=15, logoff_count=15, session_duration_avg=0.07	INSIDER	HIGH

#	Scenario Name	Description	Key Features	Label	Severity
6	Remote Work From Home	Employee consistently logs in from home VPN 8:30 AM-5:30 PM during approved remote period	remote_logon_count DAILY, logon_count=1, session_duration_avg=9	LEGITIMATE	LOW
7	Account Sharing Detected	Single account logged in from two geographically distant IPs simultaneously (UK and India)	unique_pc_count=2, logon_count=2, remote_logon_count=2	INSIDER	CRITICAL
8	Failed Login Brute Force	800 failed logon attempts in 10 minutes on one account, followed by successful access	logon_count EXTREME, session_duration_avg NEAR_ZERO, logoff_count HIGH	INSIDER	CRITICAL
9	Dormant Account Reactivation	Account inactive for 90 days suddenly logs in at 2 AM and downloads large data volumes	after_hours_logon=1, logon_count=1, session_duration_max HIGH	INSIDER	CRITICAL
10	Rapid Multi-Location Logins	User logs in from London at 9:00 AM and New York at 9:45 AM — physically impossible travel	remote_logon_count=2, unique_pc_count=2, logon_count=2	INSIDER	CRITICAL

Section 2 — File Access Scenarios

#	Scenario Name	Description	Key Features	Label	Severity
1	Bulk Sensitive File Download	Employee downloads 800 files incl. salary, HR, and financial records in 2 hours	file_access_count=800, sensitive_file_access HIGH, unusual_file_access_ratio=0.9	INSIDER	CRITICAL
2	Normal Document Editing	Employee opens, edits, saves 5-10 project documents daily within authorized folder	file_access_count=8, file_write_count=5, sensitive_file_access=0	LEGITIMATE	LOW
3	Mass File Deletion Before Exit	Employee deletes 1,200 files in 48 hours before submitting resignation letter	file_delete_count=1200, file_access_after_hours=1, file_access_count HIGH	INSIDER	CRITICAL
4	Source Code Exfiltration	Developer copies entire source code repository to a local archive file late at night	file_copy_count HIGH, large_file_transfer_count=1, sensitive_file_access HIGH	INSIDER	CRITICAL
5	Routine Backup Process	Automated backup script runs at 2 AM copying files to backup server — approved	file_copy_count HIGH, file_access_after_hours HIGH, large_file_transfer_count HIGH	LEGITIMATE	LOW
6	Anomalous File Access Pattern	Employee accesses 10x more files than their 90-day baseline, across many departments	unusual_file_access_ratio HIGH, file_access_count SPIKE, sensitive_file_access SPIKE	SUSPICIOUS	HIGH
7	Log File Tampering	User deletes system log and audit trail files — attempted cover-up of previous actions	file_delete_count HIGH, sensitive_file_access=1, file_access_after_hours=1	INSIDER	CRITICAL

#	Scenario Name	Description	Key Features	Label	Severity
8	Credential File Hunting	User searches and opens password.txt, keys.pem, .env files across multiple directories	sensitive_file_access=8, file_access_count MODERATE, unusual_file_access_ratio HIGH	INSIDER	CRITICAL
9	Ransomware Encryption Activity	All user files suddenly encrypted simultaneously — thousands of write operations in seconds	file_write_count=EXTREME, file_access_count=EXTREME, large_file_transfer_count HIGH	INSIDER	CRITICAL
10	New Employee Onboarding	New hire opens various documentation files in first week to understand company systems	file_access_count MODERATE, file_open_count MODERATE, sensitive_file_access=0	LEGITIMATE	LOW

Section 3 — USB / Device Scenarios

#	Scenario Name	Description	Key Features	Label	Severity
1	USB Data Exfiltration	Employee connects personal USB at 10 PM, copies 8 GB of files in 45 minutes, disconnects	usb_connect_count=1, usb_usage_duration=45, large_usb_transfer=1, after_hours_usb_usage=1	INSIDER	CRITICAL
2	Repeated Short USB Connects	Employee connects/disconnects USB 12 times in one hour, each lasting 2-3 minutes	usb_connect_count=12, usb_disconnect_count=12, usb_usage_duration=30	INSIDER	HIGH

#	Scenario Name	Description	Key Features	Label	Severity
3	Approved USB Data Transfer	Finance employee uses company-approved encrypted USB to transfer approved data to auditors	usb_connect_count=1, usb_usage_duration=15, large_usb_transfer=0, after_hours_usb_usage=0	LEGITIMATE	LOW
4	Multiple USBs Same Session	Employee connects 5 different USB devices in one session — copying to multiple drives	unique_usb_devices=5, usb_connect_count=5, usb_usage_duration HIGH, large_usb_transfer=1	INSIDER	CRITICAL
5	Charging Personal Phone	Employee charges personal phone via USB port — no data transfer capability on port	usb_connect_count=1, usb_usage_duration=60, usb_file_transfer_count=0, large_usb_transfer=0	LEGITIMATE	LOW
6	USB Malware Drop	Employee inserts unknown USB found in parking lot — malware auto-executes on connect	usb_connect_count=1, usb_usage_duration=2, usb_file_transfer_count=1	INSIDER	CRITICAL
7	High-Volume Sequential Copying	User connects USB and copies every folder on shared drive — 30 GB over 2 hours	large_usb_transfer=1, usb_usage_duration=120, usb_file_transfer_count HIGH, after_hours_usb_usage=1	INSIDER	CRITICAL
8	Terminated User USB Access	USB activity detected on a computer belonging to a recently terminated employee's account	after_hours_usb_usage=1, usb_connect_count=1, large_usb_transfer=1	INSIDER	CRITICAL
9	USB Use on Restricted System	Employee connects USB on air-gapped development server that prohibits all external media	usb_connect_count=1, after_hours_usb_usage=0, usb_usage_duration SHORT, unique_usb_devices=1	INSIDER	CRITICAL

Section 4 — HTTP Browsing Scenarios

#	Scenario Name	Description	Key Features	Label	Severity
1	Normal Work Browsing	Employee browses GitHub, Stack Overflow, company docs, and news sites during work hours	http_count=30, unique_url_count=25, suspicious_url_count=0, job_search_site_visits=0	LEGITIMATE	LOW
2	Cloud Storage Exfiltration	Employee uploads 4 GB of files to personal Dropbox and Google Drive after sensitive doc access	file_sharing_site_visits=8, upload_count=4, http_count MODERATE, http_after_hours=1	INSIDER	CRITICAL
3	Research on Dark Web Tools	Employee visits Tor download page, VPN services, and anonymization tools during work hours	suspicious_url_count=5, http_count MODERATE, unique_url_count=5, http_after_hours=0	SUSPICIOUS	HIGH
4	Personal Email File Forwarding	Employee accesses Gmail and Yahoo Mail 10 times, attaches company files each visit	file_sharing_site_visits=10, upload_count=10, http_count MODERATE, http_after_hours=1	INSIDER	CRITICAL
5	Malware C2 Communication	Browser silently beacons to unknown IP every 5 minutes — likely command and control traffic	http_count=100+, unique_url_count=1, suspicious_url_count=1, http_after_hours=1	INSIDER	CRITICAL
6	Online Training Course	Employee accesses Coursera and LinkedIn Learning for approved skills development training	http_count=40, unique_url_count=20, job_search_site_visits=2, suspicious_url_count=0	LEGITIMATE	LOW

#	Scenario Name	Description	Key Features	Label	Severity
7	Excessive Personal Shopping	Employee spends 4 hours on Amazon/eBay during work hours — policy violation but not a threat	http_count=80, unique_url_count=50, suspicious_url_count=0, job_search_site_visits=0	LEGITIMATE	LOW
8	Tor Network Access Attempt	Employee downloads and runs Tor browser from official site, then browses anonymously	suspicious_url_count=2, download_count=1, http_after_hours=1, unique_url_count=5	SUSPICIOUS	HIGH
9	Social Media During Work Hours	Employee spends 2 hours on Twitter/X, Instagram, Facebook — against policy, not a security threat	http_count=60, unique_url_count=20, suspicious_url_count=0, job_search_site_visits=0	LEGITIMATE	LOW
10	Phishing Site Visit	Employee clicks phishing link in email, visits fake login page, enters credentials	suspicious_url_count=1, http_count=1, download_count=0, unique_url_count=1	INSIDER	CRITICAL

Section 5 — ML Feature Impact Table

Key features used to distinguish Insider vs. Legitimate activity in ML models

LOGON Features

ID	Feature Name	Normal Values	Insider Values	ML Importance	Key Insight
L1	logon_count	1-2/day	10-15/day	HIGH	Spike = repeated access attempts
L2	logoff_count	Mirrors logon	Mismatch	MEDIUM	Mismatch = ghost sessions
L3	after_hours_logon	0	3-8/day	VERY HIGH	Strongest logon signal
L4	weekend_logon	0	1-3	HIGH	Unusual pattern, high signal
L5	unique_pc_count	1	3-8	HIGH	Lateral movement indicator
L6	session_duration_avg	7-9 hrs	0.5-2 hrs	MEDIUM	Short = data grab sessions
L7	session_duration_max	10 hrs	18+ hrs	HIGH	Extended exfil session
L8	first_logon_time	8-9 AM	0-6 AM	HIGH	Middle-of-night access
L9	last_logoff_time	17-18	23-2	MEDIUM	Late logout = after-hours work
L10	remote_logon_count	1 fixed location	Variable	HIGH	Multiple locations = stolen creds

FILE Features

ID	Feature Name	Normal Values	Insider Values	ML Importance	Key Insight
F1	file_access_count	10-20	200-500+	VERY HIGH	Bulk access = exfiltration
F2	file_open_count	Proportional	Spike	MEDIUM	Baseline comparison
F3	file_copy_count	0-2	50-200+	VERY HIGH	Direct exfil signal
F4	file_write_count	5-10	HIGH or ZERO	MEDIUM	ZERO = read-only recon

ID	Feature Name	Normal Values	Insider Values	ML Importance	Key Insight
F5	file_delete_count	0	100-2000	HIGH	Cover tracks / sabotage
F6	sensitive_file_access	0	5-50+	VERY HIGH	#1 file signal
F7	external_drive_file_copy	0	HIGH	VERY HIGH	Physical exfiltration
F8	file_access_after_hours	0	HIGH	HIGH	Night-time file access
F9	large_file_transfer_count	0-1	5-20	HIGH	DB dumps, zip archives
F10	unusual_file_access_ratio	<0.05	>0.5	VERY HIGH	Sensitive/total ratio

USB / DEVICE Features

ID	Feature Name	Normal Values	Insider Values	ML Importance	Key Insight
D1	usb_connect_count	0-1	3-10	HIGH	Multiple removable media
D2	usb_disconnect_count	Matches connect	Mismatch	LOW	Mismatch = forced removal
D3	usb_usage_duration	5-15 min	60-480 min	HIGH	Long = bulk transfer
D4	usb_file_transfer_count	1-5	100+	VERY HIGH	Estimated files moved
D5	large_usb_transfer	0	1 (flag)	VERY HIGH	Binary flag, high weight
D6	first_usb_usage_time	9-17	0-7 or 20-23	HIGH	Time of day signal
D7	after_hours_usb_usage	0	1 (flag)	VERY HIGH	Night USB = exfiltration
D8	unique_usb_devices	1	3-5	HIGH	Multiple drives

HTTP / BROWSING Features

ID	Feature Name	Normal Values	Insider Values	ML Importance	Key Insight
H1	http_count	20-50	100-500	MEDIUM	Volume alone insufficient
H2	unique_url_count	Proportional	Spike	LOW	Context feature
H3	suspicious_url_count	0	1-10	VERY HIGH	Pastebin, Tor, dark sites
H4	file_sharing_site_visits	0-1	5-20	HIGH	Dropbox, Drive uploads
H5	job_search_site_visits	0	5-30	HIGH	Departure intent signal
H6	download_count	2-5	50-200	MEDIUM	Mass download = staging
H7	upload_count	0-2	10-50	HIGH	Web-based exfiltration
H8	http_after_hours	0-2	20-100	HIGH	Night web activity